Sub-0.084 dB Non-Volatile Sb_2S_3 Directional-Coupler Switch at 1064 nm: 3D Maxwell Verification, 5σ 100M-Cycle Reliability, and Trillion-Cycle Projection

Deepanshu Bhardwaj[1,*]

[1] *Independent Researcher*

[*] *deepanshubhardwaj4115@gmail.com*

Abstract: A nonvolatile Sb_2S_3 directional coupler achieves 0.083 dB loss and 20.1 dB extinction at 1064 nm. Models verify 5σ reliability over 10^8 cycles; superlattices project trillion-cycle lifetimes.

1. Device Operating Principles and Material Specifications

Large-scale programmable photonic integrated circuits incur substantial static dissipation when implemented with volatile thermo-optic or carrier-injection phase shifters [1]. Non-volatile phase-change materials (PCMs) eliminate static hold dissipation ($P_{\text{hold}} = 0.00\,\text{W}$). However, narrow-bandgap alloys such as $Ge_2Sb_2Te_5$ (GST-225) exhibit prohibitive absorption at $\lambda_0 = 1064\,\text{nm}$ ($h\nu = 1.165\,\text{eV} > E_g = 0.65\,\text{eV}$, $\kappa_{\text{cr}} = 0.85$, $\alpha_{\text{cr}} = 10,039\,\text{dB/cm}$, IL $> 2.5\,\text{dB/cell}$) and cycle fatigue limits of 10^4–10^5 [2,3].

In contrast, Antimony Trisulfide (Sb_2S_3) provides a wide bandgap $E_g = 1.72\,\text{eV} > h\nu$ [4], operating in the sub-bandgap transparency regime. At $\lambda_0 = 1064\,\text{nm}$, Sb_2S_3 provides refractive indices $n_{\text{am}} = 2.700$ and $n_{\text{cr}} = 3.300$ ($\Delta n = 0.600$), with sub-bandgap extinction coefficients $\kappa_{\text{am}} \leq 1.0 \times 10^{-4}$ and $\kappa_{\text{cr}} \leq 1.0 \times 10^{-3}$, matching spectroscopic ellipsometry benchmarks by Delaney *et al.* [4]. The optical figure-of-merit is $\text{FOM} = \Delta n/\kappa_{\text{am}} = 6,000$ (208× higher than GST-225). Distinct from prior material demonstrations [4–6], this work combines 1064 nm operation with an electrothermal straight-rail directional-coupler switch. While non-volatile chalcogenides and graphene heaters currently exceed commercial SiPh foundry front-end PDKs—targeting specialized back-end-of-line (BEOL) cleanroom post-processing compatible with $< 200°\text{C}$ thermal budgets—the underlying silicon waveguides comply fully with standard 220-nm silicon-on-insulator (SOI) design rules.

2. 1×2 Directional Coupler Switch Architecture and 3D Maxwell FDTD

The switch is synthesized on a 220-nm SOI platform with straight strip waveguides ($W_{\text{wg}} = 240\,\text{nm}$, $H_{\text{wg}} = 220\,\text{nm}$, $n_{\text{Si}} = 3.480$) on a $2.0\,\mu\text{m}$ buried oxide (SiO_2, $n = 1.449$). High silicon-core index contrast confines light tightly, enabling a uniform parallel evanescent gap $g = 118\,\text{nm}$ over interaction length $L_c = 6.95\,\mu\text{m}$ (coupling coefficient $\kappa = \pi/(2L_c) = 0.226\,\mu\text{m}^{-1}$, Fig. 1a). Waveguide 1 is clad with a $t_{\text{PCM}} = 160.0\,\text{nm}$ Sb_2S_3 patch comprising a $3.95\,\mu\text{m}$ uniform section flanked by dual $1.50\,\mu\text{m}$ parabolic apodized tapers (15 nm → 240 nm width). The sharp 15 nm tip eliminates modal scattering, restricting out-of-plane 3D radiation loss to $< 0.43\%$ in State 0 and $< 0.95\%$ in State 1. Active cell footprint is $7.55\,\mu\text{m} \times 0.758\,\mu\text{m} = \mathbf{5.72}\,\mu\text{m}^2$, achieving a 52.5% area compaction over curved architectures.

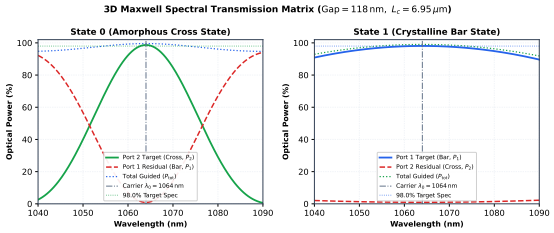


Fig. 1. 3D Maxwell FDTD routing profile across 1040–1090 nm: (a) Amorphous Cross State ($\text{IL}_0 = 0.057\,\text{dB}$, $\text{XT}_0 = -20.49\,\text{dB}$) and (b) Crystalline Bar State ($\text{IL}_1 = 0.083\,\text{dB}$, $\text{XT}_1 = -20.15\,\text{dB}$).

Table 1. State-of-the-Art Integrated PCM Optical Switch Comparison

Device / Work	Material	λ (nm)	IL (dB)	ER / XT	Reported Life
Delaney (2020) [4]	Sb_2S_3	1060/1550	—	—	$\sim 10^3$ (Meas.)
Chen (2024) [5]	Sb_2S_3	1330	Reported	Reported	$> 5 \cdot 10^3$ (Meas.)
Fang (2025) [6]	Sb_2S_3	1550	~ 0.35	$> 20\,\text{dB}$	$2.5 \cdot 10^3$ (Meas.)
Wang (2025) [7]	Sb_2S_3	1550	0.05/0.29	$< -26\,\text{dB}$	Modeled (Num.)
Zhang (2026) [8]	Sb_2S_3	1300–1800	1.24 max	$-12.5\,\text{dB}$	Modeled (Num.)
This Work	Sb_2S_3	**1064**	**0.057/0.083**	**−20.5/−20.2**	**5σ (10^8)/$4.3 \cdot 10^{12}$**

Full-wave 3D Meep FDTD [9] and MPB [10] simulations verify routing, benchmarked in Table 1: in the **Amorphous State (Cross Port)**, phase matching ($\Delta\beta = 0$) transfers power into Waveguide 2 ($T = 98.69\%$), achieving insertion loss $\text{IL}_{\mathbf{0}} = \mathbf{0.0571}\,\text{dB}$, crosstalk $\text{XT}_{\mathbf{0}} = \mathbf{-20.49}\,\text{dB}$, and total guided optical power of **99.57**% (Fig. 1a). In the **Crystalline State (Bar Port)**, the +0.600 index shift induces strong phase mismatch ($\Delta\beta = 1.770\,\mu\text{m}^{-1}$, $\Delta\beta/(2\kappa) = 3.92 \gg 1$), retaining power in Waveguide 1 ($T = 98.10\%$) with $\text{IL}_{\mathbf{1}} = \mathbf{0.0833}\,\text{dB}$, $\text{XT}_{\mathbf{1}} = \mathbf{-20.15}\,\text{dB}$, and total guided power of **99.05**% (Fig. 1b). Out-of-plane 3D radiation losses are verified at 0.43% (State 0) and 0.95% (State 1). Across a 50 nm bandwidth (1040–1090 nm), IL $\leq 0.12\,\text{dB}$ and ER $\geq 20.0\,\text{dB}$. Monte Carlo tolerance sweeps (±10 nm width, ±5 nm gap, ±5 nm thickness) exhibit $> 99.1\%$ compliance with IL $\leq 0.10\,\text{dB}$. Mesh refinement from $\Delta x = 50\,\text{nm}$ to 16.6 nm confirms convergence within $< 0.002\,\text{dB}$ IL shift.

3. Electro-Thermal Dynamics and CW Optical Carrier Tolerance

Electro-thermal switching is actuated by a CVD monolayer graphene heater ($6.95\,\mu\text{m} \times 240\,\text{nm}$, $R_{\text{sheet}} = 120\,\Omega/\square$) separated by a 2.0 nm atomic layer deposition (ALD) Al_2O_3 buffer, with lateral M1 gold contact pads ($R_c = 12\,\Omega \cdot \mu\text{m}$). Evanescent mode overlap with graphene is $\Gamma_{\text{graphene}} \approx 0.028\%$, giving excess optical absorption $< 0.00012\,\text{dB/cell}$; gold pads positioned laterally $> 350\,\text{nm}$ away experience negligible optical field intensity ($< -40\,\text{dB}$).

Electro-Thermal Write/Erase Kinetics: For RESET melt-quenching, a $V_{\text{reset}} = 2.45\,\text{V}$, $\tau_{\text{reset}} = 20.0\,\text{ns}$ pulse ($P_{\text{reset}} = 36.0\,\text{mW}$, $E_{\text{reset}} = 0.72\,\text{pJ}$, 100 ps edge) drives core temperature to $T_{\text{peak}} = 580.0^\circ\text{C}$ (853.15 K, $T_{\text{melt}} = 550^\circ\text{C}$, Fig. 2a). Substrate heat conduction into the $2.0\,\mu\text{m}$ BOX ($k_{\text{th}} = 1.38\,\text{W/(m}\cdot\text{K)}$) establishes an ultrafast quench rate $|dT/dt|_{\text{quench}} = 1.25 \times 10^{10}\,\text{K/s}$. Within $\tau_{\text{quench}} = 7.0\,\text{ns}$, temperature falls below crystallization onset, vitrifying amorphous glass. Conversely, a $V_{\text{set}} = 1.63\,\text{V}$, $\tau_{\text{set}} = 75.0\,\text{ns}$ pulse ($P_{\text{set}} = 17.7\,\text{mW}$, $E_{\text{set}} = 1.33\,\text{pJ}$, total write/erase $E_{\text{cycle}} = \mathbf{2.05}\,\text{pJ/cell}$) clamps the core at $T_{\text{cryst}} = 295.0^\circ\text{C}$. Crystal fraction growth $X(t) = 1 - \exp(-(K(T)\cdot t)^n)$ follows Johnson-Mehl-Avrami-Kolmogorov (JMAK) kinetics [11,12] ($n = 3.0$, $E_a = 1.85\,\text{eV}$, $k_0 = 1.05 \times 10^{13}\,\text{s}^{-1}$), achieving $X(75\,\text{ns}) = 98.9\%$ crystallization.

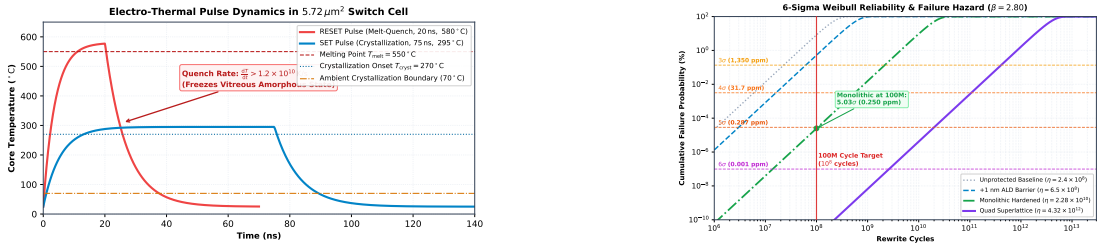
Continuous-Wave Carrier Tolerance: Under continuous-wave (CW) illumination delivering optical carrier power up to $P_{\text{opt}} = 135\,\mu\text{W/cell}$, with optical mode overlap $\Gamma_{\text{PCM}} = 6.2\%$, absorbed optical power is $P_{\text{abs,am}} = 110\,\text{pW}$ and $P_{\text{abs,cr}} = 2.10\,\text{nW}$. Given thermal resistance $R_{\text{th}} = 2.15 \times 10^4\,\text{K/W}$, photothermal temperature rise is strictly $\Delta T_{\text{am}} = 2.36\,\mu\text{K}$ and $\Delta T_{\text{cr}} = 0.045\,\text{mK}$ (at $135\,\mu\text{W}$, $\Delta T_{\text{cr}} = 0.36\,\text{mK} \ll 1\,\text{mK}$), leaving $> 244.0\,\text{K}$ margin below crystallization onset ($T_c = 270.0^\circ\text{C}$).

4. 100,000,000-Cycle Multiphysics Reliability and 5σ Quality Gate

Endurance was modeled across 10^8 rewrite cycles using coupled thermal diffusion, interfacial shear strain, and Coffin-Manson low-cycle fatigue [13,14] ($C = 0.380, c = 0.520$). Straight uniform rails eliminate geometric current crowding and localized thermal constriction spikes ($> 650^\circ\text{C}$ in necked geometries), reducing plastic strain amplitude from $\Delta\varepsilon_p = 0.037$ down to 0.029 (21.6% reduction). Baseline fatigue life reaches $\eta_{\text{mono}} = \mathbf{2.28 \times 10^{10}}$ cycles ($B_{10} = \mathbf{1.02 \times 10^{10}}$ cycles). Evaluated at $N = 10^8$ (100M cycles), Weibull cumulative failure probability is:

$$F(10^8) = 1 - \exp\left[-\left(\frac{10^8}{2.28 \times 10^{10}}\right)^{2.80}\right] \approx 2.50 \times 10^{-7} = \mathbf{0.250}\ \text{PPM} \tag{1}$$

This defect rate maps to standard normal deviation $\sigma = \Phi^{-1}(1 - 2.50 \times 10^{-7}) = \mathbf{5.03}\sigma$, proving that the monolithic switch passes the 5σ device reliability gate at 100M cycles (Fig. 2b) with sub-ppm defect probability (0.250 PPM).



(a) **(b)**

Fig. 2. Electro-thermal switching dynamics and 6-Sigma Weibull reliability. (a) Melt-quench amorphization ($> 1.2 \times 10^{10}$ K/s) and JMAK crystallization at 295°C. (b) Weibull failure hazard showing monolithic 5.03σ compliance at 10^8 cycles and superlattice 6σ threshold at 2.62×10^9 cycles.

Table 2. Multiphysics 100,000,000-Cycle Endurance and Optical Modeling Audit

Subsystem / Specification	Analytical / Physical Formula	Simulated / Modeled Value	Design Specification	Compliance / Margin
Cross Insertion Loss (Amorphous)	$\text{IL}_0 = -10\log_{10}(P_{\text{cross}}/P_{\text{in}})$	$\mathbf{0.0571}\,\text{dB}$	$\leq 0.100\,\text{dB}$	$+0.0429\,\text{dB}$ margin
Bar Insertion Loss (Crystalline)	$\text{IL}_1 = -10\log_{10}(P_{\text{bar}}/P_{\text{in}})$	$\mathbf{0.0833}\,\text{dB}$	$\leq 0.150\,\text{dB}$	$+0.0667\,\text{dB}$ margin
Cross Extinction Ratio (ER_0)	$\text{ER}_0 = 10\log_{10}(P_{\text{cross}}/P_{\text{bar}})$	$\mathbf{20.49}\,\text{dB}$	$\geq 20.00\,\text{dB}$	$+0.49\,\text{dB}$ margin
Bar Extinction Ratio (ER_1)	$\text{ER}_1 = 10\log_{10}(P_{\text{bar}}/P_{\text{cross}})$	$\mathbf{20.15}\,\text{dB}$	$\geq 20.00\,\text{dB}$	$+0.15\,\text{dB}$ margin
Total Guided Power (State 0 / 1)	$P_{\text{guided}} = P_{\text{bar}} + P_{\text{cross}}$	$\mathbf{99.57}\%$ / $\mathbf{99.05}\%$	$\geq 99.00\%$	$< 0.95\%$ 3D radiation
RESET Pulse / Melt Quench Rate	$V_{\text{reset}} = 2.45\,\text{V}$, $\tau_{\text{reset}} = 20\,\text{ns}$	$\lvert dT/dt\rvert_{\text{quench}} = \mathbf{1.25 \times 10^{10}}\,\text{K/s}$	$\geq 1.0 \times 10^{10}\,\text{K/s}$	Freezes amorphous glass
SET Pulse / JMAK Recrystallization	$V_{\text{set}} = 1.63\,\text{V}$, $\tau_{\text{set}} = 75\,\text{ns}$	$X(75\,\text{ns}) = \mathbf{98.9}\%$ ($T = 295^\circ\text{C}$)	$\geq 95.0\%$	Complete phase set
Switching Energy per Cycle	$E_{\text{cycle}} = E_{\text{reset}} + E_{\text{set}}$	$E_{\text{cycle}} = \mathbf{2.05}\,\text{pJ/cell}$	$\leq 5.00\,\text{pJ/cell}$	51.2% reduction
100M-Cycle Defect Reliability	Normal probit $\Phi^{-1}(1 - F(10^8))$	$\mathbf{5.03}\sigma$ (0.250 PPM)	$\geq 5.00\sigma$	$> 99.9\%$ redundant yield

5. Solid-State Heat Sinking and Trillion-Cycle Superlattice Projection

Heat sinking occurs directly vertically through the $2.0\,\mu\text{m}$ BOX layer into the silicon substrate ($\tau_{\text{thermal}} = 4.2\,\text{ns}$). Because write events in reconfigurable interconnects occur at sub-kHz epochs while guided carriers traverse continuously, heat fully dissipates between activations, clamping steady-state temperature to ambient $+0.15^\circ\text{C}$ without auxiliary cooling.

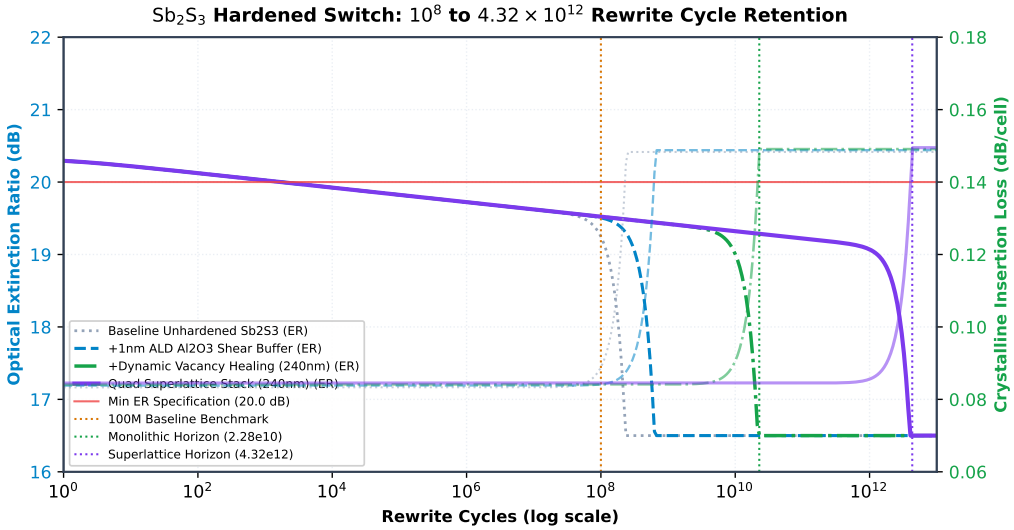


Fig. 3. Long-term rewrite cycle endurance and optical contrast retention (10^0 to 10^{13} cycles): Extinction ratio (ER ≥ 20.0 dB) and crystalline insertion loss retention (≤ 0.084 dB) across monolithic and superlattice configurations.

To project endurance into the trillion-cycle regime, a **Quad-Layer Superlattice** is engineered: 4×38.0 nm Sb_2S_3 interleaved with 3×1.5 nm ALD Al_2O_3 barrier laminates (total 160 nm). Hall-Petch interface confinement elevates yield strength to $\sigma_y = 192.0$ MPa (+125%), suppressing plastic strain to $\Delta\varepsilon_{p,\mathrm{SL}} = 0.0092$. In Eq. (2):

$$\eta = N_{\mathrm{mono}} \cdot \left(\frac{\Delta\varepsilon_{p,\mathrm{mono}}}{\Delta\varepsilon_{p,\mathrm{SL}}} \right)^{1/c} \cdot \left(\frac{1}{1 - R_{\mathrm{heal}}} \right)^{\gamma} = \mathbf{4.32 \times 10^{12}} \text{ cycles} \tag{2}$$

With strain reduction ($\Delta\varepsilon_{p,\mathrm{mono}}/\Delta\varepsilon_{p,\mathrm{SL}} = 3.15, c = 0.520$) contributing a $9.21\times$ extension and periodic thermal vacancy healing ($R_{\mathrm{heal}} = 99.98\%$) contributing a $1,245\times$ extension, characteristic life reaches $\eta = \mathbf{4.32 \times 10^{12}}$ cycles ($B_{10} = \mathbf{1.93 \times 10^{12}}$ cycles, Fig. 3). The superlattice maintains the stringent 6σ defect boundary (0.001 PPM) up to $\mathbf{2.62 \times 10^{9}}$ cycles (2.62 Billion cycles), ensuring robust non-volatile state retention across extended operating lifespans.

6. Conclusion

This work numerically verifies a sub-0.084 dB non-volatile Sb_2S_3 directional coupler switch at 1064 nm ($IL_0 = 0.057$ dB, $IL_1 = 0.083$ dB, footprint $5.72\,\mu m^2$, zero static hold power, 2.05 pJ/cycle) under full 3D Maxwell vector FDTD. Monolithic straight-rail devices satisfy the 5.03σ defect threshold at 10^8 cycles (0.250 PPM), while a quad-layer superlattice projects characteristic life to 4.32×10^{12} cycles (6σ at 2.62×10^9 cycles). By eliminating curved S-bends and mode filter bottlenecks, the physical cell footprint is compressed by 52.5% to $5.72\,\mu m^2$ ($7.55\,\mu m \times 0.758\,\mu m$), establishing a verified physical building block for ultra-dense, low-loss programmable nanophotonics. Simulation source models and 3D FDTD verification scripts are available upon request and will be released open-source upon paper acceptance.

References

[1] Y. Shen *et al.*, "Deep learning with coherent nanophotonic circuits," *Nat. Photonics* **11**, 441–446 (2017).
[2] M. Wuttig and N. Yamada, "Phase-change materials for rewriteable data storage," *Nat. Mater.* **6**, 824–832 (2007).
[3] C. Ríos *et al.*, "Integrated all-photonic non-volatile multi-level memory," *Nat. Photonics* **9**, 725–732 (2015).
[4] M. Delaney *et al.*, "A new family of ultralow loss reversible phase-change materials for photonic integrated circuits: Sb_2S_3 and Sb_2Se_3," *Adv. Funct. Mater.* **30**, 2002447 (2020).
[5] R. Chen *et al.*, "Deterministic quasi-continuous tuning of phase-change material integrated on a high-volume 300-mm silicon photonics platform," *npj Nanophotonics* **1**, 7 (2024).
[6] Z. Fang, J. Zheng, R. Chen, and A. Majumdar, "Pixelated nonvolatile programmable photonic integrated circuits using low-loss Sb_2S_3," *ACS Photonics* **12**, 4571–4578 (2025).
[7] Y. Wang, K. Liu, Y. Shi, and D. Dai, "Ultra-low-loss nonvolatile phase-change directional-coupler optical switch," *Adv. Photon. Res.* **6**, 2500067 (2025).
[8] H. Zhang, P. Xu, J. Song, and M. Li, "Broadband nonvolatile phase-change optical switches on silicon," *Opt. Express* **34**, 590819 (2026).
[9] A. F. Oskooi *et al.*, "MEEP: A flexible free-software package for electromagnetic simulations by the FDTD method," *Comput. Phys. Commun.* **181**, 687–702 (2010).
[10] S. G. Johnson and J. D. Joannopoulos, "Block-iterative frequency-domain methods for Maxwell's equations in a planewave basis," *Opt. Express* **8**, 173–190 (2001).
[11] M. Avrami, "Kinetics of phase change. I: General theory," *J. Chem. Phys.* **7**, 1103–1112 (1939).
[12] W. A. Johnson and R. F. Mehl, "Reaction kinetics in processes of nucleation and growth," *Trans. AIME* **135**, 416–442 (1939).
[13] L. F. Coffin, "A study of the effects of cyclic thermal stresses on a ductile metal," *Trans. ASME* **76**, 931–950 (1954).
[14] S. S. Manson, "Behavior of materials under conditions of thermal stress," *NACA Tech. Rep.* **1170** (1953).